

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — MASTER TOPICAL PRACTICE PACK: 13A

CHEMICAL EQUILIBRIA (K_c, K_p, FACTORS & TOTAL ENTROPY RELATION)

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 13: CHEMICAL EQUILIBRIA
SUB-TOPIC:	13A: Chemical Equilibria (K _c , K _p , Factors & Total Entropy Relation)
TOTAL QUESTIONS:	50 Authentic Past Paper Questions (Tier 1 & Tier 2)
TOTAL MARKS:	229 Marks
EXAMINER FAQs:	10 Empirical Examiner Traps & Model Answers Included

HOW TO USE THIS A* TOPICAL PACK:

1. Tier 1 (Q1–Q25): Mid-Easy to Moderate foundational questions (1–3 marks, MCQs, core calculations).
2. Tier 2 (Q26–Q50): Hard & A* Challenge questions (4–6 marks, multi-step calculations, multi-spectral NMR, extended answers).
3. Review the complete worked mark scheme in Section C after attempting each tier.
4. Master the 10 Empirical Examiner FAQs in Section D to eliminate common candidate traps.

SECTION A: TIER 1 PRACTICE QUESTIONS (FOUNDATIONAL TO MODERATE) — Q1 to Q25**Q1. 1. Equilibrium Constant Kc Expression & Units****[3 Marks]**
WCH14/01/Jan23/Q9For the reversible reaction: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$

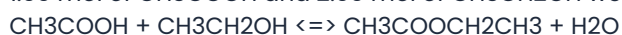
- (a) Write the expression for the equilibrium constant, Kc. [1]

- (b) Deduce the units of Kc for this reaction. [2]

Q2. 2. Homogeneous vs Heterogeneous Equilibria Kc Rules**[4 Marks]**
WCH14/01/Oct22/Q10Consider the heterogeneous equilibrium: $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$

- (a) Explain why solid
- CaCO_3
- and
- CaO
- are omitted from the Kc expression. [2]

- (b) Write the simplified Kc expression and state its units. [2]

Q3. 3. ICE Table Calculation for Kc**[5 Marks]**
WCH14/01/Jun22/Q111.00 mol of CH_3COOH and 2.00 mol of $\text{CH}_3\text{CH}_2\text{OH}$ were reacted in a 1.00 dm³ vessel:

At equilibrium, 0.80 mol of ester was formed.

- (a) Calculate the equilibrium amounts of
- CH_3COOH
- ,
- $\text{CH}_3\text{CH}_2\text{OH}$
- , and
- H_2O
- . [3]

(b) Calculate K_c for esterification. [2]

Q4. 4. Equilibrium Constant K_p Expression & Partial Pressures

[4 Marks]

WCH14/01/Jan22/Q12

For $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ at total pressure P_{total} .

(a) Define mole fraction x_A and partial pressure p_A . [2]

(b) Write the expression for K_p and deduce its units. [2]

Q5. 5. Equilibrium Pressure Calculation for K_p

[5 Marks]

WCH14/01/Oct21/Q11

In a mixture of N_2O_4 and NO_2 at 298 K, total pressure = 2.00 atm.

Mole fraction of N_2O_4 = 0.40.

(a) Calculate the partial pressures of N_2O_4 and NO_2 . [2]

(b) Write K_p for $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ and calculate its value with units. [3]

Q6. 6. Effect of Temperature on Kc and Kp for Exothermic Reactions**[4 Marks]****WCH14/01/Jun21/Q12**

For $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ ($\Delta H = -92.2 \text{ kJ mol}^{-1}$).

(a) Explain why increasing temperature decreases the value of Kc. [3]

(b) Relate this to Le Chatelier's principle. [1]

Q7. 7. Effect of Temperature on K for Endothermic Reactions**[3 Marks]****WCH14/01/Jan21/Q13**

For $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ ($\Delta H = +57.2 \text{ kJ mol}^{-1}$).

(a) Predict and explain how increasing temperature affects Kp. [3]

Q8. 8. Why Pressure and Concentration Changes Do NOT Change K**[4 Marks]****WCH14/01/Oct20/Q11**

For $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$, total pressure is doubled.

(a) Explain why doubling total pressure shifts equilibrium position to the right but leaves Kp unchanged. [4]

Q9. 9. Why Catalysts Do NOT Change Equilibrium Constant K**[3 Marks]****WCH14/01/Jun20/Q13**

A catalyst is added to an equilibrium mixture.

- (a) Explain why a catalyst increases the rate of achieving equilibrium but has zero effect on the value of K_c or K_p . [3]

Q10. 10. Derivation & Calculation of $\Delta S_{\text{total}} = R \ln K$ **[4 Marks]****WCH14/01/Jan20/Q14**

At thermodynamic equilibrium, total entropy change relates to K by $\Delta S_{\text{total}} = R \ln K$.

- (a) Calculate ΔS_{total} at 298 K for a reaction with $K = 1.80$ atm. [2]

- (b) Calculate K if $\Delta S_{\text{total}} = +50.0 \text{ J K}^{-1} \text{ mol}^{-1}$. [2]

Q11. 11. Degree of Dissociation Calculation for PCl_5 **[5 Marks]****WCH14/01/Oct19/Q14**

$\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$. 1.00 mol of PCl_5 was heated in a 10.0 dm³ vessel. At equilibrium, 0.40 mol of Cl_2 was present.

- (a) Calculate degree of dissociation α . [1]

- (b) Calculate K_c including units. [4]

Q12. 12. Esterification Equilibrium Calculation with Volume Cancellation**[4 Marks]**
WCH14/01/Jun19/Q15

$\text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH} \rightleftharpoons \text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O}$. $K_c = 4.00$.

(a) Explain why the container volume V cancels out in the K_c expression for this reaction. [2]

(b) Calculate moles of ester formed when 1.00 mol of acid and 1.00 mol of alcohol reach equilibrium. [2]

Q13. 13. Kp Calculation for Methanol Synthesis**[5 Marks]**
WCH14/01/Jan19/Q16

$\text{CO(g)} + 2\text{H}_2\text{(g)} \rightleftharpoons \text{CH}_3\text{OH(g)}$. Equilibrium partial pressures: $p_{\text{CO}} = 4.0 \text{ atm}$, $p_{\text{H}_2} = 8.0 \text{ atm}$, $p_{\text{CH}_3\text{OH}} = 2.0 \text{ atm}$.

(a) Write K_p expression. [1]

(b) Calculate K_p and deduce units. [3]

(c) Predict effect of increasing total pressure on methanol yield. [1]

Q14. 14. Relationship Between Kc and Kp via Ideal Gas Law**[4 Marks]****WCH14/01/sample/Q14**

$K_p = K_c (RT)^{\Delta n}$ where Δn = change in gas moles.

(a) For $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, state Δn . [1]

(b) Calculate K_p at 500 K if $K_c = 0.060 \text{ dm}^6 \text{ mol}^{-2}$. ($R = 0.0821 \text{ dm}^3 \text{ atm K}^{-1} \text{ mol}^{-1}$) [3]

Q15. 15. Heterogeneous Equilibrium Kp for Dehydration of Copper Sulfate**[3 Marks]****WCH14/01/sample/Q15**

$CuSO_4 \cdot 5H_2O(s) \rightleftharpoons CuSO_4(s) + 5H_2O(g)$.

(a) Write the K_p expression for this heterogeneous reaction. [1]

(b) If the equilibrium vapor pressure of water is 0.030 atm at 298 K, calculate K_p . [2]

Q16. 16. ICE Table with Quadratic Equation Solution**[5 Marks]****WCH14/01/sample/Q16**

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$. $K_c = 54.0$ at 700 K.

Initial: $[H_2] = 1.00 \text{ M}$, $[I_2] = 2.00 \text{ M}$, $[HI] = 0.00 \text{ M}$.

(a) Set up the K_c expression in terms of x (moles of H_2 reacted). [2]

(b) Solve for x and calculate equilibrium concentrations of H_2 , I_2 , and HI . [3]

Q17. 17. Temperature Dependence of K and Enthalpy Sign

[4 Marks]

WCH14/01/Sample/Q17

Kc for a reaction is 1.0×10^{-3} at 300 K and 4.5×10^{-2} at 400 K.

(a) Deduce whether the forward reaction is exothermic or endothermic. [2]

(b) Explain using van 't Hoff relationship $\ln(K_2/K_1) = (-\Delta H / R)(1/T_2 - 1/T_1)$. [2]

Q18. 18. Calculation of Delta H from Two-Point K Data

[5 Marks]

WCH14/01/Sample/Q18

For $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$: $K_p = 0.141 \text{ atm}$ at 298 K and $K_p = 2.67 \text{ atm}$ at 373 K.

(a) Calculate ΔH in kJ mol^{-1} using $\ln(K_{p2}/K_{p1}) = (\Delta H / R)(1/T_1 - 1/T_2)$. [5]

Q19. 19. Total Entropy at Equilibrium Boundary**[4 Marks]****WCH14/01/Sample/Q19**

Explain why $\Delta S_{\text{total}} = 0$ when a system reaches dynamic equilibrium.

- (a) Relate forward and reverse rates, entropy changes, and free energy at equilibrium. [4]

Q20. 20. Le Chatelier's Principle vs Reaction Quotient Q**[4 Marks]****WCH14/01/Sample/Q20**

For $A + B \rightleftharpoons C + D$, $K_c = 10.0$.

A mixture currently has $[A] = 0.5 \text{ M}$, $[B] = 0.5 \text{ M}$, $[C] = 2.0 \text{ M}$, $[D] = 2.0 \text{ M}$.

- (a) Calculate the reaction quotient $Q_c = [C][D]/([A][B])$. [2]

- (b) Compare Q_c with K_c and predict which direction the reaction will proceed to reach equilibrium. [2]

Q21. 21. Effect of Inert Gas Addition on K_p **[4 Marks]****WCH14/01/Sample/Q21**

Inert argon gas is added to $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ at constant total volume.

- (a) Explain why adding an inert gas at constant volume has zero effect on the position of equilibrium or K_p . [4]

Q22. 22. Effect of Inert Gas Addition at Constant Total Pressure**[4 Marks]****WCH14/01/Sample/Q22**

Inert argon gas is added to $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ at constant total pressure.

- (a) Explain why adding argon at constant pressure causes the equilibrium to shift to the right. [4]

Q23. 23. Equilibrium Yield Optimization in Contact Process**[5 Marks]****WCH14/01/Sample/Q23**

Contact Process: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ ($\Delta H = -197 \text{ kJ mol}^{-1}$).

- (a) Explain the industrial choice of conditions: 1–2 atm pressure, 450 °C, V_2O_5 catalyst. [5]

**Q24. 24. Decomposition of Ammonium Chloride Equilibrium
Kp****[4 Marks]****WCH14/01/Sample/Q24**

$\text{NH}_4\text{Cl}(\text{s}) \rightleftharpoons \text{NH}_3(\text{g}) + \text{HCl}(\text{g})$. At equilibrium at 600 K, total pressure is 4.00 atm.

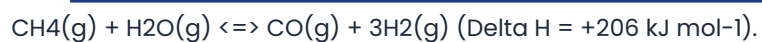
- (a) Deduce partial pressures of NH_3 and HCl . [2]

(b) Calculate K_p and give units. [2]

Q25. 25. Equilibrium Constant for Steam Reforming of Methane

[5 Marks]

WCH14/01/Sample/Q25



(a) Write K_p expression. [1]

(b) Explain why high temperature and low pressure maximize equilibrium yield of H_2 . [4]

SECTION B: TIER 2 A* CHALLENGE QUESTIONS (HARD & EXTENDED) — Q26 to Q50**Q26. 26. A* Challenge: Multi-Step ICE Table with Partial Pressures****[5 Marks]**
WCH14/01/Hard/Q26

Equilibrium: $A(g) + 2B(g) \rightleftharpoons C(g) + D(g)$.

Initial: 2.0 mol A, 3.0 mol B in a container at 10.0 atm total pressure.

At equilibrium, 1.0 mol of C is formed.

(a) Calculate equilibrium moles of A, B, C, D and total gas moles. [3]

(b) Calculate partial pressures and K_p . [2]

Q27. 27. A* Challenge: Thermodynamic Proof of Le Chatelier's Principle**[5 Marks]**
WCH14/01/Hard/Q27

Use $\Delta S_{\text{total}} = \Delta S_{\text{system}} - \Delta H / T$ to prove that increasing temperature MUST decrease K for exothermic reactions.

(a) Write the expression for $\ln K$ as a function of $1/T$. [2]

(b) Differentiate $\ln K$ w.r.t $1/T$ (van 't Hoff equation $d(\ln K)/d(1/T) = -\Delta H / R$) and explain the slope. [3]

Q28. 28. A* Challenge: Calculation of Equilibrium Composition from K_c **[5 Marks]**
WCH14/01/Hard/Q28

$$\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g}). K_c = 0.36 \text{ mol dm}^{-3} \text{ at } 373 \text{ K}.$$

If 1.00 mol of pure N_2O_4 is placed in a 2.00 dm³ flask.

(a) Calculate the equilibrium concentration of NO_2 and N_2O_4 . [5]

Q29. 29. A* Challenge: Simultaneous Chemical Equilibria Analysis

[5 Marks]
WCH14/01/Hard/Q29

Consider two coupled reactions:

Reaction 1: $A \rightleftharpoons B$ ($K_1 = 100$)

Reaction 2: $B \rightleftharpoons C$ ($K_2 = 50$)

(a) Deduce the equilibrium constant K_3 for overall reaction $A \rightleftharpoons C$. [2]

(b) If starting with 1.0 M A, calculate the equilibrium concentrations of A, B, and C. [3]

Q30. 30. A* Challenge: Total Pressure Effect on Dissociation Fraction

[5 Marks]
WCH14/01/Hard/Q30

Show that for $A(g) \rightleftharpoons 2B(g)$, the degree of dissociation α relates to total pressure P by $\alpha = \sqrt{K_p / (K_p + 4P)}$.

(a) Derive this formula starting from initial 1 mole of A. [5]

Q31. 31. A* Challenge: Temperature Dependence of K_p and Enthalpy Profile**[5 Marks]**
WCH14/01/Hard/Q31

The equilibrium constant K_p for $2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$ is 6.82 atm⁻¹ at 298 K and 0.144 atm⁻¹ at 373 K.

(a) Calculate ΔH° for dimerization of NO₂. [4]

(b) State whether dimerization is exothermic or endothermic. [1]

Q32. 32. A* Challenge: Equilibrium Constant Units in Terms of SI Units**[4 Marks]**
WCH14/01/Hard/Q32

For $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, K_p is expressed in Pa⁻².

(a) Convert K_p = 3.56 × 10⁻⁵ atm⁻² into Pa⁻² (1 atm = 101325 Pa). [2]

(b) Express Pa in SI base units (kg, m, s). [2]

Q33. 33. A* Challenge: Gas Phase Decomposition of HI Kinetic vs Equilibrium Control**[5 Marks]**
WCH14/01/Hard/Q33

$2\text{HI}(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + \text{I}_2(\text{g})$ ($\Delta H = +12.5 \text{ kJ mol}^{-1}$). $K_c = 0.020$ at 700 K.

(a) Calculate the percentage of HI decomposed at equilibrium. [3]

(b) Explain why increasing temperature increases both rate of reaching equilibrium and equilibrium yield of H_2 . [2]

Q34. 34. A* Challenge: Water Gas Shift Reaction Equilibrium**[5 Marks]**
WCH14/01/Hard/Q34

$\text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{H}_2(\text{g})$ ($\Delta H = -41.2 \text{ kJ mol}^{-1}$). $K_c = 1.00$ at 1100 K.

(a) If starting with 1.00 mol CO and 2.00 mol H_2O , calculate equilibrium composition. [3]

(b) Explain why pressure has no effect on equilibrium composition. [2]

Q35. 35. A* Challenge: Relate K_p to Gibbs Free Energy Change ΔG° **[5 Marks]**
WCH14/01/Hard/Q35

$\Delta G^\circ = -RT \ln K_p$.

(a) Calculate ΔG° at 298 K for $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$ where $K_p = 0.141 \text{ atm}$. [3]

(b) State what a positive ΔG° indicates about standard state feasibility. [2]

Q36. 36. A* Challenge: Dissociation of Dinitrogen Tetroxide at Variable Volumes

[5 Marks]
WCH14/01/Hard/Q36

$\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$. $K_c = 0.0046 \text{ mol dm}^{-3}$ at 298 K.

(a) Calculate degree of dissociation α when 1.00 mol N_2O_4 is in 10.0 dm³ vs 100.0 dm³. [4]

(b) Explain Ostwald's dilution effect on gaseous dissociation. [1]

Q37. 37. A* Challenge: Synthesis of Phosgene COCl_2 Equilibrium K_p

[5 Marks]
WCH14/01/Hard/Q37

$\text{CO}(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{COCl}_2(\text{g})$. $K_p = 15.0 \text{ atm}^{-1}$ at 373 K.

Equilibrium partial pressure of COCl_2 is 3.00 atm, and $p_{\text{CO}} = p_{\text{Cl}_2}$.

(a) Calculate the partial pressure of CO at equilibrium. [3]

(b) Calculate total pressure of the equilibrium mixture. [2]

**Q38. 38. A* Challenge: Homogeneous Gas Phase Equilibrium
Calculation for SO₃ Formation**

[5 Marks]
WCH14/01/Hard/Q38

$2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$. $K_c = 280 \text{ dm}^3 \text{ mol}^{-1}$ at 1000 K.

In a 2.00 dm³ vessel, equilibrium mixture contains 0.40 mol SO₂ and 0.20 mol O₂.

(a) Calculate the concentration of SO₃ at equilibrium. [3]

(b) Calculate total moles of SO₃ present in the 2.00 dm³ vessel. [2]

**Q39. 39. A* Challenge: Ammonia Decomposition K_p and
Partial Pressure Ratios**

[5 Marks]
WCH14/01/Hard/Q39

$2\text{NH}_3(\text{g}) \rightleftharpoons \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$. Pure NH₃ is introduced at initial pressure P₀.

(a) Express K_p in terms of equilibrium partial pressure p_{NH₃} and mole fraction ratio. [3]

(b) If 20% of NH₃ decomposes at total pressure 10.0 atm, calculate K_p. [2]

Q40. 40. A* Challenge: Le Chatelier Pressure Paradox in Gas Phase Reactions**[5 Marks]**
WCH14/01/Hard/Q40

For $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$, increasing pressure increases $[\text{NO}_2]$ in mol dm^{-3} but decreases mole fraction of NO_2 .

(a) Explain this apparent paradox using Boyle's law and equilibrium shift. [5]

Q41. 41. A* Challenge: Calculation of K_c for Ester Hydrolysis with Added Acid Catalyst**[5 Marks]**
WCH14/01/Hard/Q41

1.00 mol ethyl ethanoate + 1.00 mol water + 0.10 mol $\text{HCl}(\text{aq})$ catalyst (containing water).

Equilibrium mixture titrated: 0.40 mol ester remains at equilibrium.

(a) Account for water contributed by HCl solution in the ICE table. [2]

(b) Calculate K_c . [3]

Q42. 42. A* Challenge: Equilibrium Constant Kp of Steam Methane Reforming**[5 Marks]**
WCH14/01/Hard/Q42

$\text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + 3\text{H}_2(\text{g})$. At 1000 K, total pressure = 5.00 atm, $p_{\text{CH}_4} = 0.50$ atm, $p_{\text{H}_2\text{O}} = 0.50$ atm.

(a) Calculate p_{CO} and p_{H_2} given $p_{\text{H}_2} = 3 p_{\text{CO}}$. [2]

(b) Calculate K_p . [3]

Q43. 43. A* Challenge: Temperature Dependence of Solvation Equilibria**[5 Marks]**
WCH14/01/Hard/Q43

Gas solubility equilibrium: $\text{O}_2(\text{g}) \rightleftharpoons \text{O}_2(\text{aq})$ ($\Delta H = -12.0 \text{ kJ mol}^{-1}$).

(a) Explain using thermodynamics why dissolved oxygen concentration in water decreases as water temperature rises. [3]

(b) Discuss environmental implications for aquatic life in thermal pollution. [2]

Q44. 44. A* Challenge: Multi-Gas Equilibrium Partitioning in Closed Container**[5 Marks]**
WCH14/01/Hard/Q44

$\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$ ($K_c = 1.0 \times 10^{-5}$ at 1500 K).

Air mixture initially: 0.80 M N_2 , 0.20 M O_2 .

(a) Calculate equilibrium concentration of NO. [3]

(b) Why can the approximation $(0.80 - x) \sim 0.80$ be used? [2]

Q45. 45. A* Challenge: Van 't Hoff Isochore Derivation & Graph Analysis

[5 Marks]
WCH14/01/Hard/Q45

A plot of $\ln K$ against $1/T$ for a reversible reaction yields a straight line with gradient $+4500 \text{ K}$ and y -intercept -12.5 .

(a) Calculate ΔH° in kJ mol^{-1} . [2]

(b) Calculate $\Delta S_{\text{system}}^\circ$ in $\text{J K}^{-1} \text{ mol}^{-1}$. [3]

Q46. 46. A* Challenge: Gas Phase Equilibrium Dissociation of N_2O_4 at High Altitude

[5 Marks]
WCH14/01/Hard/Q46

At sea level ($P = 1.0 \text{ atm}$), N_2O_4 is 20% dissociated. At high altitude ($P = 0.50 \text{ atm}$).

(a) Calculate K_p at 298 K . [2]

(b) Calculate degree of dissociation α at 0.50 atm . [3]

Q47. 47. A* Challenge: Industrial Synthesis of Hydrogen Cyanide Equilibrium**[5 Marks]**
WCH14/01/Hard/Q47

Andrussov process: $\text{CH}_4(\text{g}) + \text{NH}_3(\text{g}) + 1.5 \text{O}_2(\text{g}) \rightarrow \text{HCN}(\text{g}) + 3\text{H}_2\text{O}(\text{g})$ ($\Delta H = -473 \text{ kJ mol}^{-1}$).

(a) Explain how temperature and pressure affect equilibrium yield vs rate. [3]

(b) Why is Pt-Rh gauze catalyst used at 1100 °C? [2]

Q48. 48. A* Challenge: Equilibrium Constant of Carbonate Buffer System**[5 Marks]**
WCH14/01/Hard/Q48

In ocean chemistry: $\text{CO}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_2\text{CO}_3(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{HCO}_3^-(\text{aq})$.

(a) Explain how increasing atmospheric CO_2 partial pressure affects ocean pH (ocean acidification). [3]

(b) Relate this to carbonate ion dissolution $\text{CaCO}_3(\text{s}) + \text{H}^+(\text{aq}) \rightarrow \text{Ca}^{2+}(\text{aq}) + \text{HCO}_3^-(\text{aq})$. [2]

Q49. 49. A* Challenge: Equilibrium Analysis of Dehydration of Hydrated Cobalt Chloride**[5 Marks]****WCH14/01/Hard/Q49**

$\text{CoCl}_2 \cdot 6\text{H}_2\text{O}(\text{s})$ (pink) $\rightleftharpoons$ $\text{CoCl}_2(\text{s})$ (blue) + $6\text{H}_2\text{O}(\text{g})$ ($\Delta H = +350 \text{ kJ mol}^{-1}$).

(a) Explain why cobalt chloride paper turns blue in dry air and pink in humid air. [3]

(b) Write $K_p = (p_{\text{H}_2\text{O}})^6$. [2]

Q50. 50. A* Challenge: Complete Equilibria & Thermodynamics Master Synthesis**[6 Marks]****WCH14/01/Hard/Q50**

Synthesis of phosgene: $\text{CO}(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{COCl}_2(\text{g})$

$\Delta H^\circ = -108 \text{ kJ mol}^{-1}$, $\Delta S_{\text{system}}^\circ = -132 \text{ J K}^{-1} \text{ mol}^{-1}$.

(a) Calculate ΔS_{total} and K_p at 298 K. [4]

(b) Calculate the temperature above which phosgene spontaneously dissociates into CO and Cl_2 . [2]

SECTION C: WORKED MARK SCHEME (Q1–Q50) — 13A

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. Equilibrium Constant K_c Expression & Units 1. (a) $K_c = [\text{NH}_3]^2 / ([\text{N}_2][\text{H}_2]^3)$ (1). 1. (b) Units = $(\text{mol dm}^{-3})^2 / ((\text{mol dm}^{-3})(\text{mol dm}^{-3})^3) = 1 / (\text{mol dm}^{-3})^2 = \text{dm}^6 \text{ mol}^{-2}$ (2).	[3]
Q2	Q2: 2. Homogeneous vs Heterogeneous Equilibria K_c Rules 2. (a) Concentrations of pure solids are constant and incorporated into the value of K_c (2). 2. (b) $K_c = [\text{CO}_2(\text{g})]$ (1); units = mol dm^{-3} (1).	[4]
Q3	Q3: 3. ICE Table Calculation for K_c 3. (a) $\text{CH}_3\text{COOH eq} = 1.00 - 0.80 = 0.20 \text{ mol}$ (1). $\text{CH}_3\text{CH}_2\text{OH eq} = 2.00 - 0.80 = 1.20 \text{ mol}$ (1). $\text{H}_2\text{O eq} = 0.80 \text{ mol}$ (1). 3. (b) $K_c = (0.80 \times 0.80) / (0.20 \times 1.20) = 0.64 / 0.24 = 2.67$ (dimensionless) (2).	[5]
Q4	Q4: 4. Equilibrium Constant K_p Expression & Partial Pressures 4. (a) Mole fraction $x_A = \text{moles A} / \text{total gas moles}$ (1). Partial pressure $p_A = x_A \times P_{\text{total}}$ (1). 4. (b) $K_p = (p_{\text{SO}_3})^2 / ((p_{\text{SO}_2})^2 (p_{\text{O}_2}))$ (1); units = atm^{-1} or kPa^{-1} or Pa^{-1} (1).	[4]
Q5	Q5: 5. Equilibrium Pressure Calculation for K_p 5. (a) $p_{\text{N}_2\text{O}_4} = 0.40 \times 2.00 = 0.80 \text{ atm}$ (1). $x_{\text{NO}_2} = 1.0 - 0.40 = 0.60 \Rightarrow p_{\text{NO}_2} = 0.60 \times 2.00 = 1.20 \text{ atm}$ (1). 5. (b) $K_p = (p_{\text{NO}_2})^2 / p_{\text{N}_2\text{O}_4} = (1.20)^2 / 0.80 = 1.44 / 0.80 = 1.80 \text{ atm}$ (3).	[5]
Q6	Q6: 6. Effect of Temperature on K_c and K_p for Exothermic Reactions 6. (a) Increasing temperature favours the endothermic reverse reaction to absorb heat (1). Concentration of reactants increases and products decreases (1). Since $K_c = [\text{NH}_3]^2 / ([\text{N}_2][\text{H}_2]^3)$, K_c decreases (1). 6. (b) Le Chatelier's principle states equilibrium shifts to oppose temperature increase (1).	[4]
Q7	Q7: 7. Effect of Temperature on K for Endothermic Reactions 7. (a) Forward reaction is endothermic ($\Delta H > 0$) (1). Increasing temperature shifts equilibrium to the right to absorb heat (1). Partial pressure of NO_2 increases relative to N_2O_4 , so K_p increases (1).	[3]
Q8	Q8: 8. Why Pressure and Concentration Changes Do NOT Change K 8. (a) Doubling pressure initially doubles all partial pressures (1). The denominator $(p_{\text{SO}_2})^2(p_{\text{O}_2})$ increases by $2^3 = 8x$, while numerator $(p_{\text{SO}_3})^2$ increases by $2^2 = 4x$ (1). Quotient $Q_p < K_p$ (1). System responds by converting $2\text{SO}_2 + \text{O}_2$ into 2SO_3 until partial pressure ratio restores exact value of K_p (1).	[4]
Q9	Q9: 9. Why Catalysts Do NOT Change Equilibrium Constant K 9. (a) Catalyst speeds up both forward and reverse reaction rates by the exact same factor (1). Lowers activation energy E_a for both directions equally (1). Ratio of rate constants $k_{\text{forward}} / k_{\text{reverse}} = K$ remains constant (1).	[3]
Q10	Q10: 10. Derivation & Calculation of $\Delta S_{\text{total}} = R \ln K$ 10. (a) $\Delta S_{\text{total}} = 8.31 \times \ln(1.80) = 8.31 \times 0.5878 = +4.88 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). 10. (b) $\ln K = 50.0 / 8.31 = 6.0169 \Rightarrow K = e^{6.0169} = 410.3$ (2).	[4]
Q11	Q11: 11. Degree of Dissociation Calculation for PCl_5 11. (a) $\alpha = 0.40 / 1.00 = 0.40$ (40%) (1). 11. (b) $[\text{PCl}_5]_{\text{eq}} = 0.60/10 = 0.060 \text{ M}$; $[\text{PCl}_3]_{\text{eq}} = 0.40/10 = 0.040 \text{ M}$; $[\text{Cl}_2]_{\text{eq}} = 0.040 \text{ M}$ (2). $K_c = (0.040 \times 0.040) / 0.060 = 0.00267 \text{ mol dm}^{-3}$ (2).	[5]
Q12	Q12: 12. Esterification Equilibrium Calculation with Volume Cancellation 12. (a) Equal number of moles of reactants (2 mol) and products (2 mol), so volume V in numerator and denominator cancels completely (2). 12. (b) $K_c = x^2 / (1-x)^2 = 4.00 \Rightarrow x / (1-x) = 2.00 \Rightarrow x = 2 - 2x \Rightarrow 3x = 2 \Rightarrow x = 0.67 \text{ mol}$ (2).	[4]

Q13	Q13: 13. Kp Calculation for Methanol Synthesis 13. (a) $K_p = p_{\text{CH}_3\text{OH}} / (p_{\text{CO}} \times (p_{\text{H}_2})^2)$ (1). 13. (b) $K_p = 2.0 / (4.0 \times 64.0) = 2.0 / 256.0 = 7.81 \times 10^{-3} \text{ atm}^{-2}$ (3). 13. (c) Yield increases (3 gas moles form 1 gas mole) (1).	[5]
Q14	Q14: 14. Relationship Between Kc and Kp via Ideal Gas Law 14. (a) $\Delta n = 2 - 4 = -2$ (1). 14. (b) $K_p = K_c (RT)^{\Delta n} = 0.060 \times (0.0821 \times 500)^{-2} = 0.060 \times (41.05)^{-2} = 0.060 / 1685.1 = 3.56 \times 10^{-5} \text{ atm}^{-2}$ (3).	[4]
Q15	Q15: 15. Heterogeneous Equilibrium Kp for Dehydration of Copper Sulfate 15. (a) $K_p = (p_{\text{H}_2\text{O}})^5$ (1). 15. (b) $K_p = (0.030)^5 = 2.43 \times 10^{-8} \text{ atm}^5$ (2).	[3]
Q16	Q16: 16. ICE Table with Quadratic Equation Solution 16. (a) $K_c = (2x)^2 / ((1-x)(2-x)) = 54.0 \Rightarrow 4x^2 = 54.0(2 - 3x + x^2) \Rightarrow 50.0x^2 - 162.0x + 108.0 = 0$ (2). 16. (b) $x = 0.93 \text{ mol dm}^{-3}$ (3). $[\text{H}_2] = 0.07 \text{ M}$, $[\text{I}_2] = 1.07 \text{ M}$, $[\text{HI}] = 1.86 \text{ M}$.	[5]
Q17	Q17: 17. Temperature Dependence of K and Enthalpy Sign 17. (a) Endothermic ($\Delta H > 0$) because Kc increases as temperature increases (2). 17. (b) $\ln(K_2/K_1) > 0$ and $(1/T_2 - 1/T_1) < 0$, so $-\Delta H / R$ must be negative $\Rightarrow \Delta H > 0$ (2).	[4]
Q18	Q18: 18. Calculation of Delta H from Two-Point K Data 18. (a) $\ln(2.67 / 0.141) = \ln(18.936) = 2.941$ (1). $(1/298 - 1/373) = 6.749 \times 10^{-4} \text{ K}^{-1}$ (1). $\Delta H / R = 2.941 / 6.749 \times 10^{-4} = 4358 \text{ K}$ (2). $\Delta H = 4358 \times 8.31 = 36215 \text{ J mol}^{-1} = +36.2 \text{ kJ mol}^{-1}$ (1).	[5]
Q19	Q19: 19. Total Entropy at Equilibrium Boundary 19. (a) At dynamic equilibrium, rate of forward reaction equals rate of reverse reaction (1). Net concentration of all species remains constant (1). Universe entropy is maximized for the system composition, so $\Delta S_{\text{total}} = 0$ (1). $\Delta G = -T \Delta S_{\text{total}} = 0$ (1).	[4]
Q20	Q20: 20. Le Chatelier's Principle vs Reaction Quotient Q 20. (a) $Q_c = (2.0 \times 2.0) / (0.5 \times 0.5) = 4.0 / 0.25 = 16.0$ (2). 20. (b) $Q_c (16.0) > K_c (10.0)$ (1). Reaction will shift to the left (reverse direction) to consume C and D and form more A and B until $Q_c = K_c$ (1).	[4]
Q21	Q21: 21. Effect of Inert Gas Addition on Kp 21. (a) Adding argon increases total pressure but total volume is constant (1). Partial pressures of N ₂ , H ₂ , and NH ₃ remain completely unchanged (2). Therefore Kp and equilibrium position remain unchanged (1).	[4]
Q22	Q22: 22. Effect of Inert Gas Addition at Constant Total Pressure 22. (a) Adding argon increases total gas volume to keep total pressure constant (1). Mole fractions and partial pressures of reactant N ₂ O ₄ and product NO ₂ both decrease (1). Since $K_p = (p_{\text{NO}_2})^2 / p_{\text{N}_2\text{O}_4}$, denominator decreases less than numerator (1). System shifts right (towards more gas moles) to restore Kp (1).	[4]
Q23	Q23: 23. Equilibrium Yield Optimization in Contact Process 23. (a) Moderate pressure (1–2 atm) gives >99% yield without high cost of high-pressure equipment (2). 450 °C provides acceptable rate while keeping exothermic Kp high enough for high yield (2). V ₂ O ₅ catalyst speeds up rate without affecting yield (1).	[5]
Q24	Q24: 24. Decomposition of Ammonium Chloride Equilibrium Kp 24. (a) Equimolar gas formation $\Rightarrow p_{\text{NH}_3} = p_{\text{HCl}} = 2.00 \text{ atm}$ (2). 24. (b) $K_p = p_{\text{NH}_3} \times p_{\text{HCl}} = 2.00 \times 2.00 = 4.00 \text{ atm}^2$ (2).	[4]
Q25	Q25: 25. Equilibrium Constant for Steam Reforming of Methane 25. (a) $K_p = (p_{\text{CO}} \times (p_{\text{H}_2})^3) / (p_{\text{CH}_4} \times p_{\text{H}_2\text{O}})$ (1). 25. (b) Endothermic reaction ($\Delta H > 0$) $\Rightarrow$ high temperature shifts equilibrium right to absorb heat, increasing Kp (2). 2 gas moles form 4 gas moles $\Rightarrow$ low pressure shifts equilibrium right towards more gas moles (2).	[5]
Q26	Q26: 26. A* Challenge: Multi-Step ICE Table with Partial Pressures 26. (a) $A_{\text{eq}} = 1.0$, $B_{\text{eq}} = 1.0$, $C_{\text{eq}} = 1.0$, $D_{\text{eq}} = 1.0 \text{ mol}$. Total moles = 4.0 mol (3). 26. (b) Mole fractions all 0.25 $\Rightarrow p_A = p_B = p_C = p_D = 2.5 \text{ atm}$. $K_p = (2.5 \times 2.5) / (2.5 \times (2.5)^2) = 1 / 2.5 = 0.40 \text{ atm}^{-1}$ (2).	[5]

Q27	Q27: 27. A* Challenge: Thermodynamic Proof of Le Chatelier's Principle 27. (a) $\ln K = \Delta S_{\text{system}} / R - \Delta H / (RT)$ (2). 27. (b) Slope of $\ln K$ vs $1/T$ is $-\Delta H / R$. For exothermic reaction ($\Delta H < 0$), slope is positive. As T increases, $1/T$ decreases, so $\ln K$ decreases, meaning K decreases (3).	[5]
Q28	Q28: 28. A* Challenge: Calculation of Equilibrium Composition from K_c 28. (a) Initial $[N_2O_4] = 0.50$ M. Let $x = \text{mol dm}^{-3} N_2O_4$ reacted $\Rightarrow [N_2O_4]_{\text{eq}} = 0.50 - x$, $[NO_2]_{\text{eq}} = 2x$ (1). $K_c = (2x)^2 / (0.50 - x) = 0.36 \Rightarrow 4x^2 + 0.36x - 0.18 = 0$ (2). $x = (-0.36 + \sqrt{0.1296 + 2.88}) / 8 = (-0.36 + 1.7348) / 8 = 0.172$ M (1). $[N_2O_4]_{\text{eq}} = 0.328$ M, $[NO_2]_{\text{eq}} = 0.344$ M (1).	[5]
Q29	Q29: 29. A* Challenge: Simultaneous Chemical Equilibria Analysis 29. (a) $K_3 = K_1 \times K_2 = 100 \times 50 = 5000$ (2). 29. (b) $[B] = 100[A]$, $[C] = 50[B] = 5000[A]$. $[A] + [B] + [C] = 5001[A] = 1.0$ M $\Rightarrow [A] = 2.0 \times 10^{-4}$ M, $[B] = 0.020$ M, $[C] = 0.980$ M (3).	[5]
Q30	Q30: 30. A* Challenge: Total Pressure Effect on Dissociation Fraction 30. (a) Moles at eq: $A = 1 - \alpha$, $B = 2\alpha$. Total moles = $1 + \alpha$ (1). Partial pressures: $p_A = ((1 - \alpha) / (1 + \alpha))P$, $p_B = ((2\alpha) / (1 + \alpha))P$ (2). $K_p = (p_B)^2 / p_A = [4\alpha^2 / (1 - \alpha^2)] P$ (1). Rearranging gives $\alpha^2 = K_p / (K_p + 4P) \Rightarrow \alpha = \sqrt{K_p / (K_p + 4P)}$ (1).	[5]
Q31	Q31: 31. A* Challenge: Temperature Dependence of K_p and Enthalpy Profile 31. (a) $\ln(0.144 / 6.82) = \ln(0.02111) = -3.858$ (1). $(1/373 - 1/298) = -6.749 \times 10^{-4} \text{ K}^{-1}$ (1). $\Delta H / R = -3.858 / (-6.749 \times 10^{-4}) = 5716 \text{ K}$ (1). $\Delta H = -5716 \times 8.31 = -47.5 \text{ kJ mol}^{-1}$ (1). 31. (b) Exothermic ($\Delta H < 0$) (1).	[5]
Q32	Q32: 32. A* Challenge: Equilibrium Constant Units in Terms of SI Units 32. (a) $K_p(\text{Pa}^{-2}) = (3.56 \times 10^{-5}) / (101325)^2 = 3.47 \times 10^{-15} \text{ Pa}^{-2}$ (2). 32. (b) $\text{Pa} = \text{N m}^{-2} = (\text{kg m s}^{-2}) \text{ m}^{-2} = \text{kg m}^{-1} \text{ s}^{-2} \Rightarrow \text{Pa}^{-2} = \text{kg}^{-2} \text{ m}^2 \text{ s}^4$ (2).	[4]
Q33	Q33: 33. A* Challenge: Gas Phase Decomposition of HI Kinetic vs Equilibrium Control 33. (a) $K_c = (x/2)^2 / (1 - x)^2 = 0.020 \Rightarrow x / (2(1 - x)) = \sqrt{0.020} = 0.1414 \Rightarrow x = 0.283 - 0.283x \Rightarrow 1.283x = 0.283 \Rightarrow x = 0.221$ (22.1% decomposed) (3). 33. (b) High temp increases rate by providing $E > E_a$ for collisions (1); endothermic reaction shifts right, increasing K_c and yield (1).	[5]
Q34	Q34: 34. A* Challenge: Water Gas Shift Reaction Equilibrium 34. (a) $K_c = x^2 / ((1 - x)(2 - x)) = 1.00 \Rightarrow x^2 = 2 - 3x + x^2 \Rightarrow 3x = 2 \Rightarrow x = 0.67$ mol (3). $\text{CO} = 0.33$, $\text{H}_2\text{O} = 1.33$, $\text{CO}_2 = 0.67$, $\text{H}_2 = 0.67$ mol. 34. (b) 2 moles of gas reactants form 2 moles of gas products, so pressure changes affect numerator and denominator equally (2).	[5]
Q35	Q35: 35. A* Challenge: Relate K_p to Gibbs Free Energy Change ΔG° 35. (a) $\Delta G^\circ = -8.31 \times 298 \times \ln(0.141) = -2476.38 \times (-1.959) = +4851 \text{ J mol}^{-1} = +4.85 \text{ kJ mol}^{-1}$ (3). 35. (b) Positive ΔG° means reaction is non-spontaneous under standard conditions ($p = 1$ atm for all gases); equilibrium favors reactants (2).	[5]
Q36	Q36: 36. A* Challenge: Dissociation of Dinitrogen Tetroxide at Variable Volumes 36. (a) At 10 dm ³ : $[N_2O_4]_0 = 0.10$ M $\Rightarrow 4x^2 / (0.10 - x) = 0.0046 \Rightarrow x = 0.0102$ M ($\alpha = 10.2\%$) (2). At 100 dm ³ : $[N_2O_4]_0 = 0.010$ M $\Rightarrow 4x^2 / (0.010 - x) = 0.0046 \Rightarrow x = 0.0028$ M ($\alpha = 28.0\%$) (2). 36. (b) Increasing volume (dilution) increases degree of dissociation for reactions where gas moles increase (1).	[5]
Q37	Q37: 37. A* Challenge: Synthesis of Phosgene COCl_2 Equilibrium K_p 37. (a) $K_p = p_{\text{COCl}_2} / (p_{\text{CO}})^2 = 15.0 \Rightarrow 3.00 / (p_{\text{CO}})^2 = 15.0 \Rightarrow (p_{\text{CO}})^2 = 0.20 \Rightarrow p_{\text{CO}} = 0.447$ atm (3). 37. (b) Total pressure = $p_{\text{COCl}_2} + p_{\text{CO}} + p_{\text{Cl}_2} = 3.00 + 0.447 + 0.447 = 3.89$ atm (2).	[5]
Q38	Q38: 38. A* Challenge: Homogeneous Gas Phase Equilibrium Calculation for SO_3 Formation 38. (a) $[\text{SO}_2] = 0.20$ M, $[\text{O}_2] = 0.10$ M. $K_c = [\text{SO}_3]^2 / ((0.20)^2(0.10)) = 280 \Rightarrow [\text{SO}_3]^2 = 280 \times 0.0040 = 1.12 \Rightarrow [\text{SO}_3] = 1.058 \text{ mol dm}^{-3}$ (3). 38. (b) Moles $\text{SO}_3 = 1.058 \times 2.00 = 2.12$ mol (2).	[5]

Q39	Q39: 39. A* Challenge: Ammonia Decomposition K_p and Partial Pressure Ratios 39. (a) At eq: moles $\text{NH}_3 = 2(1-\alpha)$, $\text{N}_2 = \alpha$, $\text{H}_2 = 3\alpha$. Total = $2 + 2\alpha$ (1). $K_p = \frac{(p_{\text{N}_2} \times (p_{\text{H}_2})^3)}{(p_{\text{NH}_3})^2}$ (2). 39. (b) $\alpha = 0.20$. Moles: $\text{NH}_3 = 1.6$, $\text{N}_2 = 0.2$, $\text{H}_2 = 0.6$. Total = 2.4. Mole fracs: $\text{NH}_3=0.667$, $\text{N}_2=0.0833$, $\text{H}_2=0.25$. Partial pressures: $\text{NH}_3=6.67$, $\text{N}_2=0.833$, $\text{H}_2=2.50$. $K_p = \frac{(0.833 \times 15.625)}{44.49} = 0.292 \text{ atm}^2$ (2).	[5]
Q40	Q40: 40. A* Challenge: Le Chatelier Pressure Paradox in Gas Phase Reactions 40. (a) Compression (decreasing volume V) immediately increases concentrations of all gases (Boyle's law) (2). System responds by shifting left (towards fewer gas moles) to reduce total moles (2). Net effect: mole fraction x_{NO_2} decreases, but because volume decreased by a larger factor, equilibrium $[\text{NO}_2]$ is higher than before compression (1).	[5]
Q41	Q41: 41. A* Challenge: Calculation of K_c for Ester Hydrolysis with Added Acid Catalyst 41. (a) HCl catalyst is aqueous, adding initial water moles (2). 41. (b) Ester reacted = 0.60 mol. Acid formed = 0.60 mol, alcohol formed = 0.60 mol. Water eq = $1.00 - 0.60 = 0.40$ mol. $K_c = \frac{(0.60 \times 0.60)}{(0.40 \times 0.40)} = 0.36 / 0.16 = 2.25$ (3).	[5]
Q42	Q42: 42. A* Challenge: Equilibrium Constant K_p of Steam Methane Reforming 42. (a) $p_{\text{CO}} + p_{\text{H}_2} = 5.00 - 0.50 - 0.50 = 4.00 \text{ atm}$ (1). $4 p_{\text{CO}} = 4.00 \Rightarrow p_{\text{CO}} = 1.00 \text{ atm}$, $p_{\text{H}_2} = 3.00 \text{ atm}$ (1). 42. (b) $K_p = \frac{(1.00 \times (3.00)^3)}{(0.50 \times 0.50)} = 27.0 / 0.25 = 108 \text{ atm}^2$ (3).	[5]
Q43	Q43: 43. A* Challenge: Temperature Dependence of Solvation Equilibria 43. (a) Dissolution of O_2 gas is exothermic ($\Delta H < 0$) (1). Increasing temperature shifts equilibrium left (towards O_2 gas) to absorb heat, decreasing Henry's law solubility constant K (2). 43. (b) Warm industrial wastewater discharges reduce dissolved oxygen levels, causing asphyxiation of fish and aerobic organisms (2).	[5]
Q44	Q44: 44. A* Challenge: Multi-Gas Equilibrium Partitioning in Closed Container 44. (a) $K_c = \frac{(2x)^2}{((0.80-x)(0.20-x))} \sim \frac{4x^2}{(0.80 \times 0.20)} = \frac{4x^2}{0.16} = 1.0 \times 10^{-5}$ (1). $4x^2 = 1.6 \times 10^{-6} \Rightarrow x^2 = 4.0 \times 10^{-7} \Rightarrow x = 6.32 \times 10^{-4} \text{ M}$ (1). $[\text{NO}] = 2x = 1.26 \times 10^{-3} \text{ mol dm}^{-3}$ (1). 44. (b) K_c is extremely small (10^{-5}), so $x \ll 0.20 \text{ M}$ (approximation error $< 1\%$) (2).	[5]
Q45	Q45: 45. A* Challenge: Van 't Hoff Isochore Derivation & Graph Analysis 45. (a) Gradient = $-\Delta H / R = +4500 \text{ K} \Rightarrow \Delta H = -4500 \times 8.31 = -37395 \text{ J mol}^{-1} = -37.4 \text{ kJ mol}^{-1}$ (exothermic) (2). 45. (b) Y-intercept = $\Delta S_{\text{sys}} / R = -12.5 \Rightarrow \Delta S_{\text{sys}} = -12.5 \times 8.31 = -103.9 \text{ J K}^{-1} \text{ mol}^{-1}$ (3).	[5]
Q46	Q46: 46. A* Challenge: Gas Phase Equilibrium Dissociation of N_2O_4 at High Altitude 46. (a) $\alpha = 0.20$ at $1.0 \text{ atm} \Rightarrow K_p = \frac{4(0.20)^2}{(1 - (0.20)^2)} \times 1.0 = 0.16 / 0.96 = 0.167 \text{ atm}$ (2). 46. (b) $0.167 = \frac{4\alpha^2}{(1 - \alpha^2)} \times 0.50 \Rightarrow \frac{4\alpha^2}{(1 - \alpha^2)} = 0.3333 \Rightarrow 4.3333\alpha^2 = 0.3333 \Rightarrow \alpha^2 = 0.0769 \Rightarrow \alpha = 0.277$ (27.7% dissociated) (3).	[5]
Q47	Q47: 47. A* Challenge: Industrial Synthesis of Hydrogen Cyanide Equilibrium 47. (a) Exothermic reaction favoured by low T , but 1100°C required for high rate (1). 3.5 gas moles $\rightarrow$ 4 gas moles favoured by low pressure (2). 47. (b) Extremely high activation energy requires Pt-Rh catalyst at 1100°C for millisecond contact time (2).	[5]
Q48	Q48: 48. A* Challenge: Equilibrium Constant of Carbonate Buffer System 48. (a) Higher p_{CO_2} increases $[\text{CO}_2(\text{aq})]$, shifting equilibrium right to produce more H^+ ions, lowering ocean pH (acidification) (3). 48. (b) Increased H^+ reacts with CaCO_3 marine shells/coral, dissolving them (2).	[5]
Q49	Q49: 49. A* Challenge: Equilibrium Analysis of Dehydration of Hydrated Cobalt Chloride 49. (a) In dry air (low $p_{\text{H}_2\text{O}}$), equilibrium shifts right (towards blue CoCl_2) to replenish H_2O gas (1.5). In humid air (high $p_{\text{H}_2\text{O}}$), equilibrium shifts left (towards pink hydrate) (1.5). 49. (b) $K_p = (p_{\text{H}_2\text{O}})^6$ (2).	[5]

Q50	Q50: 50. A* Challenge: Complete Equilibria & Thermodynamics Master Synthesis 50. (a) $\Delta S_{\text{surr}} = -(-108000)/298 = +362.4 \text{ J K}^{-1} \text{ mol}^{-1}$. $\Delta S_{\text{tot}} = -132 + 362.4 = +230.4 \text{ J K}^{-1} \text{ mol}^{-1}$ (2). $\ln K_p = 230.4 / 8.31 = 27.72 \Rightarrow K_p = e^{27.72} = 1.09 \times 10^{12} \text{ atm}^{-1}$ (2). 50. (b) $T_{\text{boundary}} = \Delta H / \Delta S_{\text{sys}} = 108000 / 132 = 818.2 \text{ K}$ (545.0 °C) (2).	[6]
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SECTION D: EMPIRICAL EXAMINER FAQs & COMMON TRAPS (10 FAQs) — 13A

FAQ #1: Omitting Solids/Liquids from K_c and K_p Category: Heterogeneous
Equilibria**COMMON EXAMINER TRAP / MISTAKE:**

Including pure solids or liquids in K_c/K_p expressions.

MODEL ANSWER / EXAMINER EXPECTATION:

Concentrations of pure solids and pure liquid solvents are constant and built into the equilibrium constant value. Omit them entirely from K_c and K_p .

FAQ #2: Units of K_c and K_p Deduction

Category: Units Calculation

COMMON EXAMINER TRAP / MISTAKE:

Writing dimensionless or standard units without checking stoichiometry.

MODEL ANSWER / EXAMINER EXPECTATION:

Units depend on mole change: substitute (mol dm^{-3}) into K_c or (atm/kPa) into K_p . If gas moles on left equal gas moles on right, K is dimensionless.

FAQ #3: Effect of Pressure on K_p Value

Category: Pressure Paradox

COMMON EXAMINER TRAP / MISTAKE:

Claiming increasing total pressure increases the value of K_p .

MODEL ANSWER / EXAMINER EXPECTATION:

Total pressure changes shift the EQUILIBRIUM POSITION, but K_p itself is CONSTANT at a fixed temperature. The partial pressures adjust to maintain exact K_p .

FAQ #4: Effect of Temperature on K Sign

Category: Temperature Effect

COMMON EXAMINER TRAP / MISTAKE:

Confusing direction of K shift for exothermic vs endothermic reactions.

MODEL ANSWER / EXAMINER EXPECTATION:

For EXOTHERMIC ($\Delta H < 0$), heating DECREASES K . For ENDOTHERMIC ($\Delta H > 0$), heating INCREASES K . (Remember: K changes ONLY with temperature).

FAQ #5: Catalyst Effect on Equilibrium Yield

Category: Catalyst Fallback

COMMON EXAMINER TRAP / MISTAKE:

Stating a catalyst increases equilibrium yield or shifts position.

MODEL ANSWER / EXAMINER EXPECTATION:

A catalyst speeds up forward and reverse reactions by the EXACT SAME FACTOR. It reaches equilibrium faster but has ZERO effect on yield or K .

FAQ #6: Mole Fraction vs Partial Pressure

Category: Partial Pressure
Calculation**COMMON EXAMINER TRAP / MISTAKE:**

Using mole fraction directly in K_p instead of partial pressure.

MODEL ANSWER / EXAMINER EXPECTATION:

Partial pressure p_A = mole fraction x_A x TOTAL PRESSURE. You must multiply by total pressure before inserting into K_p .

FAQ #7: ICE Table Initial Moles vs Equilibrium Moles**Category: ICE Table Trap****COMMON EXAMINER TRAP / MISTAKE:**

Inserting initial moles directly into K_c expression without subtracting change x .

MODEL ANSWER / EXAMINER EXPECTATION:

Always construct an ICE table (Initial, Change, Equilibrium). Moles at equilibrium = Initial +/- stoichiometric change x , converted to concentration ($[] = \text{mol} / V$).

FAQ #8: Relating Total Entropy to K Formula**Category: Thermodynamic Formula****COMMON EXAMINER TRAP / MISTAKE:**

Using $\log_{10}$ instead of natural log $\ln$ in $\Delta S_{\text{total}} = R \ln K$.

MODEL ANSWER / EXAMINER EXPECTATION:

The formula uses natural logarithm ($\ln$ / base e). $\ln K = \Delta S_{\text{total}} / R \Rightarrow K = e^{(\Delta S_{\text{total}} / R)}$.
 $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$.

FAQ #9: Volume Cancellation in Kc Expressions**Category: Volume Shortcut****COMMON EXAMINER TRAP / MISTAKE:**

Dividing by volume V when gas moles on both sides are equal.

MODEL ANSWER / EXAMINER EXPECTATION:

When sum of moles of reactants equals sum of moles of products (e.g. esterification), volume V cancels out completely. K_c can be calculated directly from equilibrium moles.

FAQ #10: Reaction Quotient Q vs Equilibrium Constant K**Category: Direction of Shift****COMMON EXAMINER TRAP / MISTAKE:**

Confusing direction of shift when $Q > K$.

MODEL ANSWER / EXAMINER EXPECTATION:

If $Q < K$, system shifts RIGHT to form more products. If $Q > K$, system shifts LEFT to form more reactants until $Q = K$.